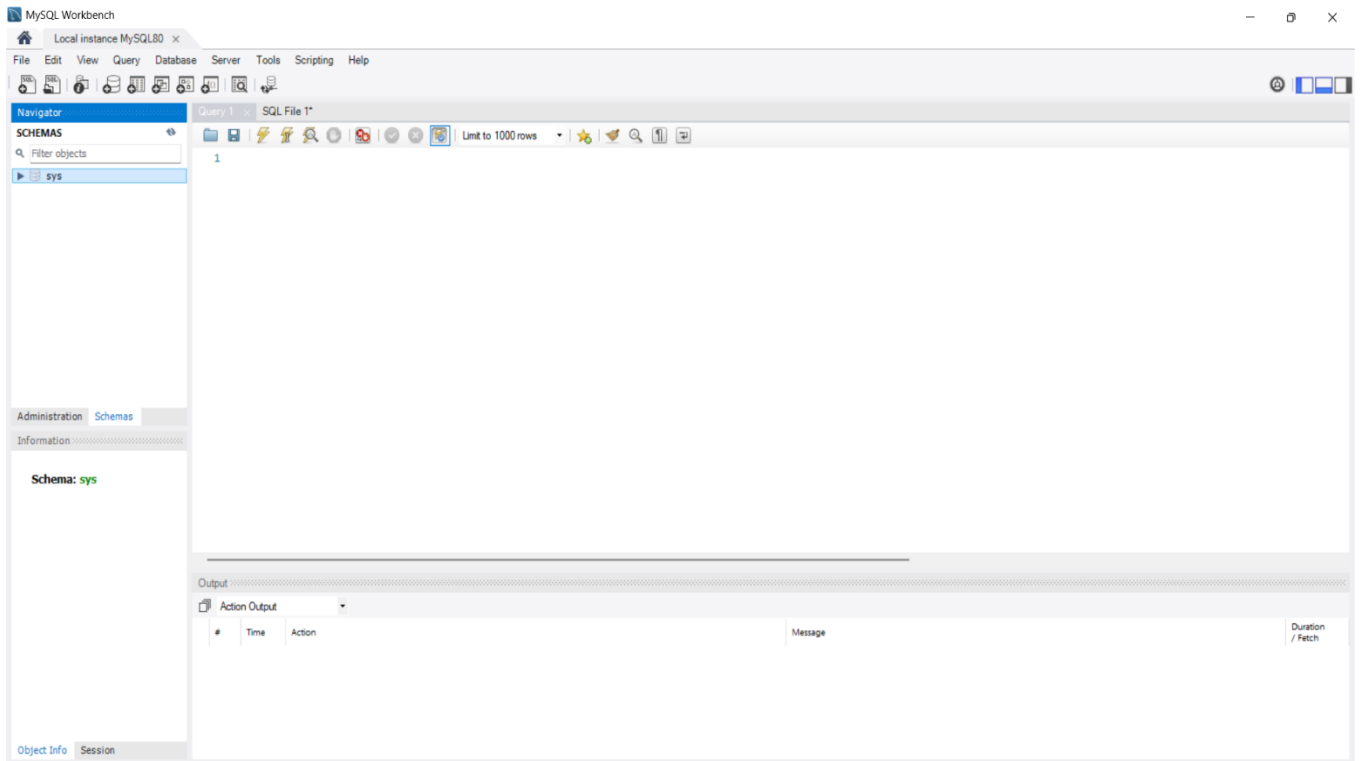


Session:-1

Question 1:-Install MySQL Workbench on your computer, connect to the local MySQL server, and take a screenshot of the default databases shown in the Schemas panel.



Question2:- Create a new database called InstaClone in MySQL Workbench, then create a table named Users with columns: user_id (INT, primary key), username (VARCHAR), email (VARCHAR), and followers_count (INT).

Ans:-

```
CREATE DATABASE InstaClone;
```

```
USE InstaClone;
```

```
CREATE TABLE Users (  
    user_id INT PRIMARY KEY,  
    username VARCHAR(100),  
    email VARCHAR(100),  
    followers_count INT);
```

followers_count INT

);

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	12:13:01	CREATE DATABASE InstaClone	1 row(s) affected	0.016 sec
✓ 2	12:13:06	USE InstaClone	0 row(s) affected	0.000 sec
✓ 3	12:13:10	CREATE TABLE Users (user_id INT PRIMARY KEY, username VARCHAR(100), email VARCHAR(100), f...	0 row(s) affected	0.031 sec

Question3 :- Insert 3 sample users into the Users table you created, using realistic Instagram-style usernames and follower counts.

Ans :-

USE InstaClone;

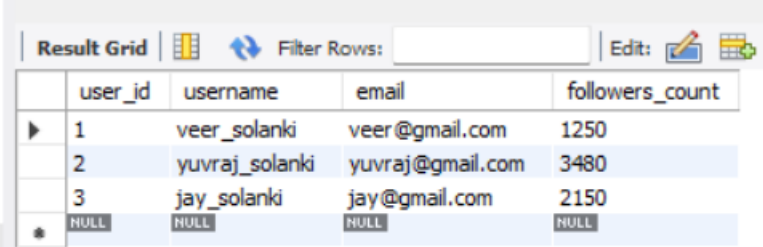
```
INSERT INTO Users (user_id, username, email, followers_count)VALUES
```

```
(1, 'veer_solanki', 'veer@gmail.com', 1250),
```

```
(2, 'yuvraj_solanki', 'yuvraj@gmail.com', 3480),
```

```
(3, 'jay_solanki', 'jay@gmail.com', 2150);
```

```
SELECT * FROM Users;
```



	user_id	username	email	followers_count
▶	1	veer_solanki	veer@gmail.com	1250
	2	yuvraj_solanki	yuvraj@gmail.com	3480
	3	jay_solanki	jay@gmail.com	2150
✱	NULL	NULL	NULL	NULL

Question4 :- Create another table in the InstaClone database called Posts with columns: post_id (INT, primary key), user_id (INT), caption (VARCHAR), and post_date (DATE). Add a foreign key from Posts.user_id to Users.user_id to establish a relationship.

Ans:-

USE InstaClone;

CREATE TABLE Posts (

post_id INT PRIMARY KEY,

user_id INT,

caption VARCHAR(255),

post_date DATE,

FOREIGN KEY (user_id) REFERENCES Users(user_id)

);



Question5 :- Write a short comparison (3-4 lines each) of MySQL,

PostgreSQL, Oracle, and SQLite, focusing on where each one is commonly used (for example: web apps, mobile apps, enterprise systems, etc.).

Ans:-

MySQL :-

MySQL is widely used for web applications and websites. It is popular with PHP, WordPress, and many backend systems. It is easy to use and suitable for small to large applications.

PostgreSQL :-

PostgreSQL is commonly used for web apps, data-intensive applications, and analytics. It supports advanced SQL features and complex data types. It is often chosen when strong data integrity and advanced database features are needed.

Oracle:-

Oracle Database is mainly used in large enterprise systems. Banks, governments, telecom companies, and large organizations commonly use it. It is designed for high performance, security, scalability, and large amounts of data.

SQLite:-

SQLite is commonly used in mobile apps, desktop applications, and embedded systems. It is lightweight and does not require a separate database server. It is especially useful when an application needs a small, local database.

